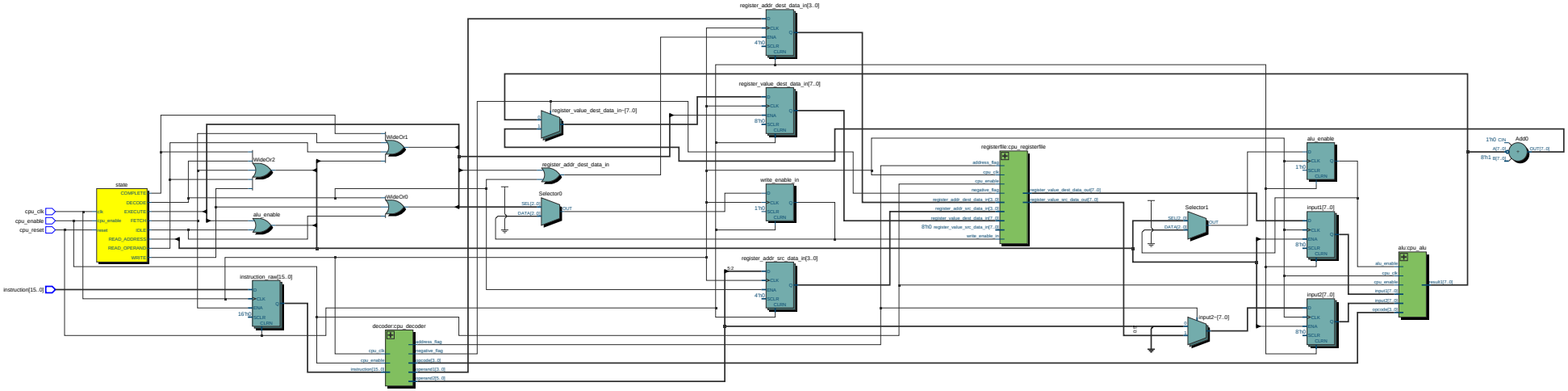
cpu_clk
cpu_enable
cpu_reset
instruction[15..0]
state
COMPLETE
DECODE
EXECUTE
FETCH
IDLE
READ_ADDRESS
READ_OPERANDS
WRITE
WideOr1
WideOr2
WideOr0
alu_enable
instruction_raw[15..0]
16'h0
decoder:cpu_decoder
register_addr_dest_data_in[3..0]
4'h0
register_value_dest_data_in[7..0]
8'h0
register_value_dest_data_in~[7..0]
register_addr_dest_data_in
Selector0
write_enable_in
1'h0
register_addr_src_data_in[3..0]
s:2
registerfile:cpu_registerfile
address_flag
cpu_clk
cpu_enable
negative_flag
register_addr_dest_data_in[3..0]
register_addr_src_data_in[3..0]
register_value_dest_data_in[7..0]
register_value_src_data_in[7..0]
write_enable_in
register_value_dest_data_out[7..0]
register_value_src_data_out[7..0]
Selector1
input2~[7..0]
alu_enable
input1[7..0]
input2[7..0]
alu:cpu_alu
result[7..0]
opcode[3..0]
17'h0
Add0
OUT[7..0]
8'h1